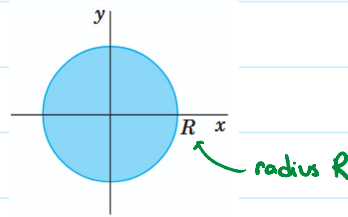


Circular Membrane

Saturday, July 26, 2025 6:21 PM

13-Laplace's Equation on a circular membrane.

13.1 Suppose our membrane is circular:



Then the 2D wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u$$

can be expressed in polar coordinates:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} \right)$$

13.2 We will massively simplify our problem by assuming:

* All vibrations are radially symmetric: i.e. $u(r, \theta, t) = u(r, t)$

$$\Rightarrow \frac{\partial u}{\partial \theta} = \frac{\partial^2 u}{\partial \theta^2} = 0$$

13.3 The problem:

$$c^2 \left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} \right) = \frac{\partial^2 u}{\partial t^2}$$

with

$$\text{BC: } u(R, t) = 0$$

$$\text{and IC: } u(r, 0) = f(r)$$

$$u_t(r, 0) = g(r), \text{ but we will only solve for } g=0$$

$$\text{i.e. } u_t(r, 0) = 0$$

13.4 Separation of variables:

$$\text{Let } u(r, t) = W(r)G(t)$$

Then

$$c^2 G(t) \left[\frac{d^2 W(r)}{dr^2} + \frac{1}{r} \frac{dW(r)}{dr} \right] = W(r) \frac{d^2 G(t)}{dt^2}$$

$$\text{i.e. } \frac{1}{W} W'' + \frac{1}{rW} W' = \frac{\ddot{G}}{G} = -k^2$$

ie. $\frac{1}{W} W'' + \frac{1}{rW} W' = \frac{\ddot{G}}{c^2 G} = -k^2$

Thus $\ddot{G} + k^2 c^2 G = 0 \Rightarrow G(t) = A \cos(\lambda t) + B \sin(\lambda t)$

By now, we know that $u_t(r, 0) = 0 \Rightarrow B = 0$

so: $G(t) = A \cos(\lambda t)$ where $\lambda = kc$

On the other hand: $\frac{1}{W} W'' + \frac{1}{rW} W' = -k^2 \Rightarrow W'' + \frac{1}{r} W' + k^2 W = 0$

13.5 The only physically viable solution to

$$\frac{d^2 W}{dr^2} + \frac{1}{r} \frac{dW}{dr} + k^2 W = 0$$

is $W = J_0(kr)$, where $J_0(\cdot)$ is the zeroth order Bessel function of the first kind

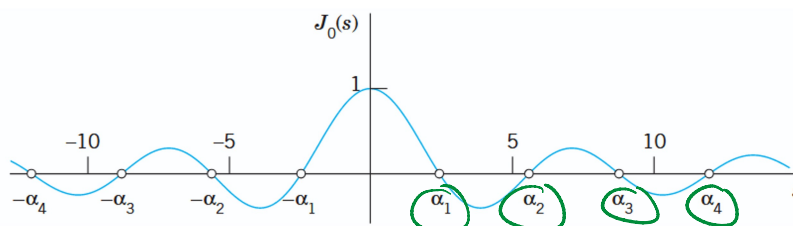


Fig. 308. Bessel function $J_0(s)$

These numbers are known

13.6 So far, we have:

$$u(r, t) = J_0(kr) \cdot A \cos(\lambda t)$$

Now, to satisfy the condition

$$u(R, t) = 0, \text{ we need } J_0(kR) = 0$$

so kR must be a root of the Bessel function $J_0(\cdot)$

in other words: $kR = \alpha_i$ for some $i = 1, 2, 3, 4, \dots$

so we say: $k_m = \alpha_m / R$, $m = 1, 2, 3, \dots$

Furthermore: $\lambda = c k \Rightarrow \lambda_m = \frac{c \alpha_m}{R}$

So we say: $u_m(r, t) = A_m \cos\left(\frac{c \alpha_m}{R} t\right) J_0\left(\frac{\alpha_m}{R} r\right)$

13.6 Finally, to satisfy $u(r, 0) = f(r)$
we insist

$$u(r, t) = \sum_{m=1}^{\infty} u_m(r, t) = \sum_{m=1}^{\infty} A_m \cos\left(\frac{c \alpha_m}{R} t\right) J_0\left(\frac{\alpha_m}{R} r\right)$$

Then $u(r, 0) = f(r)$ becomes:

$$\sum_{m=1}^{\infty} A_m J_0\left(\frac{\alpha_m}{R} r\right) = f(r)$$

These A_m are called the coefficients of the **Fourier-Bessel series**

$$A_m = \frac{2}{R^2 J_1^2(\alpha_m)} \int_0^R r f(r) J_0\left(\frac{\alpha_m}{R} r\right) dr$$

Values of α_m , $J_1(\alpha_m)$ are known and will be provided to you in a table.

The integral $\int_0^R r f(r) J_0\left(\frac{\alpha_m}{R} r\right) dr$ is usually numerically computed, but for special cases of $f(r)$ we can figure it out.

13.7 Example: Suppose

$$f(r) = 11.08 J_0\left(\frac{2.4048}{2} r\right) - 0.140 J_0\left(\frac{8.6537}{2} r\right)$$

and $u(r, t)$ satisfies $u_{tt} = c^2 \nabla^2 u$, $c = 1$

with BC: $u(2, t) = 0$

IC: $u(r, 0) = f(r)$, $u_t(r, 0) = 0$

on a circular membrane of radius $R = 2$.

m	α_m	$J_1(\alpha_m)$
1	2.40483	0.51915
2	5.52008	-0.34026
3	8.65373	0.27145
4	11.79153	-0.23246
5	14.93092	0.20655
6	18.07106	-0.18773
7	21.21164	0.17327
8	24.35247	-0.16170
9	27.49348	0.15218
10	30.63461	-0.14417

on a circular membrane of radius $R=2$.

What is $u(r, t)$?

9	27.49348	0.15218
10	30.63461	-0.14417

Soln We recognize: $f(r) = \underbrace{1.108}_{A_1} J_0\left(\frac{\alpha_1}{R} r\right) - \underbrace{0.140}_{A_3} J_0\left(\frac{\alpha_3}{R} r\right)$

So we get: $A_1 = 1.108$, $A_2 = 0$, $A_3 = -0.140$, $A_4 = A_5 = \dots = 0$

Then

$$u(r, t) = 1.108 \cos\left(\frac{c \alpha_1}{R} t\right) J_0\left(\frac{\alpha_1}{R} r\right) - 0.140 \cos\left(\frac{c \alpha_3}{R} t\right) J_0\left(\frac{\alpha_3}{R} r\right)$$

$$= 1.108 \cos\left(\frac{2.40483}{2} t\right) J_0\left(\frac{2.40483}{2} r\right) - 0.140 \cos\left(\frac{8.65373}{2} t\right) J_0\left(\frac{8.65373}{2} r\right)$$

13.8 Harmonics Frequency

In the solution for $u(r, t)$, λ_m are harmonic wavenumbers, and

$f_m = \frac{\lambda_m}{2\pi}$ are called harmonic frequencies.

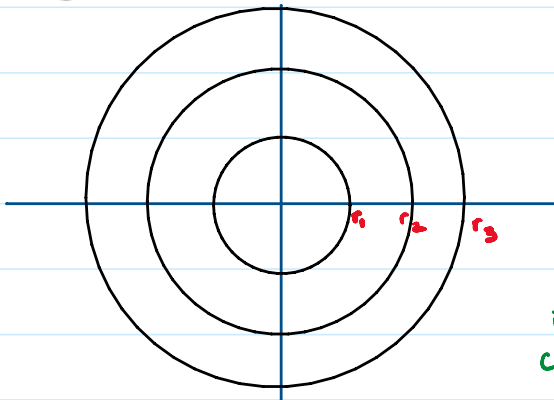
For instance, in the example above, the relevant harmonic frequencies are

$$f_1 = \frac{\lambda_1}{2\pi} = \frac{(c \alpha_1 / R)}{2\pi} = \frac{(2.40483/2)}{2\pi} \approx 0.1914 \text{ Hz}$$

$$f_3 = \frac{\lambda_3}{2\pi} = \frac{(c \alpha_3 / R)}{2\pi} = \frac{(8.65373/2)}{2\pi} \approx 0.6886 \text{ Hz}$$

13.9 Nodal Circles:

$m=3$



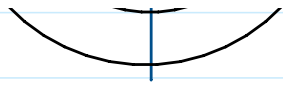
inner circles

For $J_0\left(\frac{\alpha_m}{R} r\right)$, whenever $J_0(\cdot) = 0$ we get a nodal circle.

For example, when $m=3$, there will be three nodal circles:

First one when: $\frac{\alpha_3}{R} r_1 = \alpha_1 \Rightarrow r_1 = \left(\frac{\alpha_1}{\alpha_3}\right) R$

Second one when $\frac{\alpha_3}{R} r_2 = \alpha_2 \Rightarrow r_2 = \left(\frac{\alpha_2}{\alpha_3}\right) R$



circles ↑

Second one when $\frac{\alpha_2}{R} r_2 = \alpha_2 \Rightarrow r_2 = \left(\frac{\alpha_2}{\alpha_3}\right) R$

Third one when $\frac{\alpha_3}{R} r_3 = \alpha_3 \Rightarrow r_3 = \left(\frac{\alpha_3}{\alpha_3}\right) R = R$

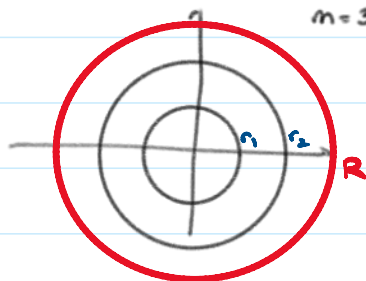
Note: the final circle will always be of radius R
when discussing nodal circles for harmonics, we only care about the inner circles.

13.10 For the example above, draw the nodal circles for all harmonics and identify the radius of each circle

Soln: We have $u(r, t) = A_1 \cos\left(\frac{\alpha_1}{R} t\right) J_0\left(\frac{\alpha_1}{R} r\right) + A_3 \cos\left(\frac{\alpha_3}{R} t\right) J_0\left(\frac{\alpha_3}{R} r\right)$

when $m=1$, there is no nodal circle for the harmonic (no inner circle)

when $m=3$, there are two nodal circles for the harmonic, with radius:



$$\text{with } r_1 = \left(\frac{\alpha_1}{\alpha_3}\right) R = \left(\frac{2.40483}{8.65373}\right)^2 \approx 0.5558$$

$$r_2 = \left(\frac{\alpha_2}{\alpha_3}\right) R = \left(\frac{5.52008}{8.65373}\right)^2 \approx 1.2758$$